SHoEbox: balanced learning through broader context

November 6, 2026 | Continuation of published checkpoint `7f3ce94`

Language and memory now train through the same graph in each optimizer update. The new control caps each source's gradients separately in the core, readout, recall and publication groups. Memory queries now follow natural-text distractors and updates to competing facts. All 312,440 parameters train; the inference architecture is unchanged.

Completed comparison

Held-out measurement	Group clipping	Direct sum
Selected update	96	128
Natural-byte NLL	3.0922	3.0814
All-answer NLL	1.9282	1.8560
Exact-answer NLL	1.7051	1.6080
Exact-answer accuracy	21.7%	26.5%
Exact NLL, long-delay documents	1.6817	1.6204
Exact NLL beyond 96 bytes	1.6808	1.6325
Exact accuracy beyond 96 bytes	22.4%	23.5%
Preceding task-test answer NLL	2.0927	2.0187

Lower NLL is better. These scores use 2,048 natural-byte targets, 288 answer targets in 16 fresh contextual documents, and a separate long-delay set with three times the configured retention gaps. Acquisition questions remain short; the beyond-96-byte rows exclude them and contain 98 exact-answer targets. The preceding task test uses its original eight documents and seed. The ordinary exact-answer subset contains 226 targets. Uniform choice among its four valid values gives 25% accuracy; exact recall remains close to that baseline even as answer NLL improves.

On these same held-out tests, the starting checkpoint scores 3.3672 language NLL, 2.0579 answer NLL, and 1.9224 long-delay exact NLL. Its exact-answer NLL beyond 96 bytes is 1.9497.

Group clipping changes language NLL by +0.0108, answer NLL by +0.0721 and long-delay exact NLL by +0.0613 relative to direct summation at independently validation-selected checkpoints. Both arms began at validation language/answer NLL of 3.4004/2.0608.

At the final update, clipping scores 3.1572 language and 1.9056 answer NLL; direct summation scores 3.0814 and 1.8560. Clipping remains an optional control alongside direct summation.

Exposure and controls

Both arms complete 128 paired updates: 16,384 ordered natural-text bytes, 16,384 task-stream bytes, 727 answer targets, and 5,505,024 expert executions per arm. Task-stream totals include their embedded natural text. Each pair uses two forward/backward passes at identical weights, then one AdamW step. The language cursor continues from the parent; optimizer and runtime states start fresh.

Batch 4, window 32 bytes, gradient segment 16, constant LR 0.001. Selection uses mean validation language and answer NLL at updates 0/32/64/96/128. Natural distractors respect train/validation/test corpus boundaries. The four-entity task uses acquisition queries, corrections, two cycles, six questions per cycle, and minimum delays of 4/48/96 bytes. Query labels and fact provenance are never model inputs. No replay store is added.

Whole-rollout controls below start from fresh test state; positive deltas mean worse answer NLL. These include changes to later computation and publication.

Intervention	Clipping delta	Sum delta
Block regional reads	+2.2562	+0.3714
Always publish	-0.0000	-0.0012
Remove receiver demand	+0.0001	+0.0003
Always commit state	+0.0000	-0.0003

Retention and gradient behavior

Producer reads at queries	Mean publication age, bytes	Older than 16 bytes
Group clipping	0.00	0.0%
Direct sum	0.16	0.0%

Age covers selected valid A/B/C/U/V publications across receivers and rounds. Publication age differs from fact age: recent publications can carry older recurrent information. Long-held publications remain rare; the next target is to make useful information survive repeated publication updates.

Group, clipping arm	Median language norm	Median memory norm	Language capped	Memory capped
core	0.928	0.423	46.1%	18.0%
readout	0.634	2.851	0.0%	99.2%
recall	0.111	0.080	1.6%	4.7%
publication	0.019	0.008	0.0%	2.3%

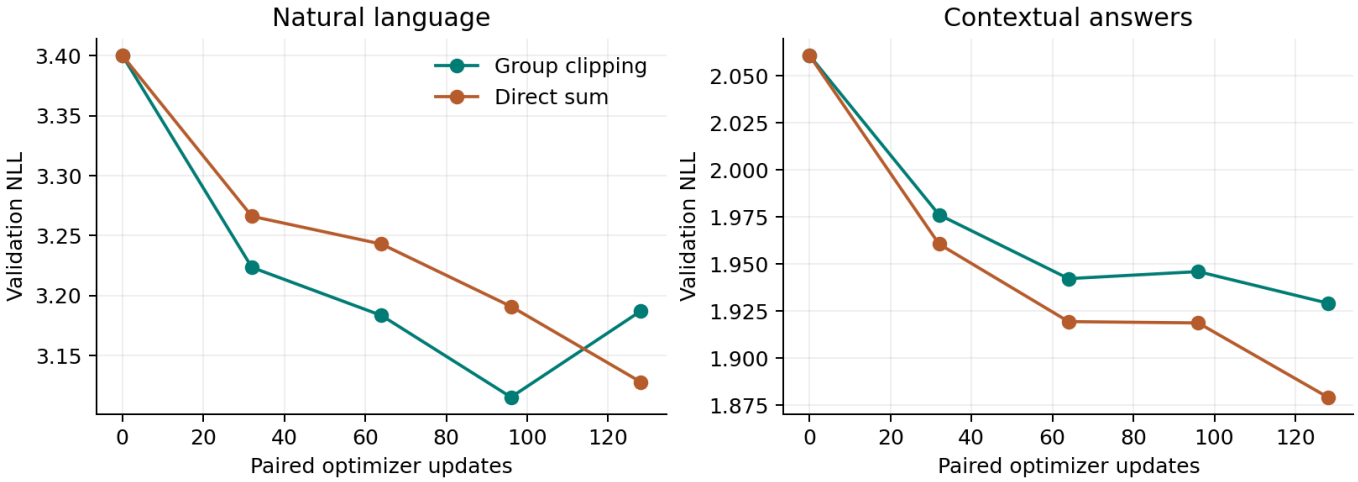
Each source/group is capped at norm 1 before addition, followed by the common global cap of 1. The direct-sum control retains the common global cap. Full logs also retain gradient cosines and delayed-credit magnitudes for both sources.

Verification and continuation

Regression verification passed 558 unique checks: 547 existing checks and the complete 11-test new suite. The driver stopped reporting during the new suite, which was then rerun separately and passed. New checks cover exact gradient summation, simultaneous source weights, independent state and credit, parent-cursor transfer, exact resume, corpus separation, target provenance, publication-age accounting and evaluation-cache invalidation.

The execution resumed from completed pairs 119 and 121 after a session interruption. Exposure totals cover the retained trajectories; the validation schedule stayed fixed.

The publication/state/demand/payload controls remain independently ablatable. Their intended coordination remains marked as a potentially threshold-dependent emergent feature. Continue with task-driven retention and shared regional learning.



Reproduce

```
python scripts/run_context_balance.py --parent artifacts/context_balance/parent.pt --parallel
python scripts/check_all.py
python scripts/evaluate_context_parent.py
python scripts/summarize_context_balance.py
python scripts/report_context_balance.py
```

The archive includes source, corpus, parent, selected/final checkpoints, optimizer/runtime/credit states, full gradient logs, query traces and hashes. Detailed protocol: `docs/context_balance.md`. Machine-readable results: `experiments/text_regions/results/context_balance/summary.json`.